1. Executive Summary (≈1,500 words)

- **Purpose**: Provide a high-level overview of the SimulaFly Admin Panel as a unified "company OS" that sits on top of the consumer app, merchant portal, wallet, support, and analytics. Emphasize that SimulaFly blends commerce and AI, requiring a dashboard that monitors both real-time operations (orders, wallets, support tickets) and AI usage/cost.
- **Core Objectives**: Summarize key goals – manage consumer and merchant support, operations, and finance in one place; monitor user and merchant activity; enforce policies (verification, billing); surface analytics on conversions and AI costs; enable targeted promotions/referrals; and maintain trust & safety.
- **Role-Based Access**: Mention that different teams (Org Manager, Ops, Sales, Finance, Support, Analytics, Technical) will see different modules. Provide a high-level role matrix (to be detailed later) mapping roles to modules (e.g. Org Manager: full access; Support: only users/orders/tickets; Finance: wallets/invoices; etc.).
- **Benchmarking**: Cite Amazon Seller Central and Flipkart as analogies – both use dashboards to show sales snapshots, order funnels, inventory and performance metrics. Mention Google Analytics style navigation for multi-layer reports (Home/Reports/Admin). Emphasize the need for real-time data (like "Today's Sales" or "Impressions" seen in Flipkart) and on-demand filters.
- **High-Level Features**: Bullet-list top modules: Dashboard, Consumer Support, Merchant Support, Orders, Merchants, Stores, Products, Wallet/Finance, Invoices, Offers/Coupons, Referral Management, Pricing Control, AI Usage, Analytics, Campaigns, Trust & Safety, System Health, Audit Logs, Settings. Each section shows what metrics and controls are present.
- **Success Metrics**: Outline the KPIs to improve (merchant activation, user activation, booking conversion, churn, AI cost efficiency, support response times).
- **Organization Structure**: Present a simplified hierarchy: Organization → Merchants → Stores → Products, with Users and AI as parallel strands. Show (in text or small mermaid) the core entities and relations.

2. Platform Architecture (≈1,200 words)

2.1 Entity Model

- Describe primary objects: **Organization** (SimulaFly itself), **Merchant**, **Store** (each merchant can have multiple stores), **Product**, **Order/Booking**, **User/Consumer**, **Wallet**, **Invoice**, **Offer/Promotion**, **Coupon**, **Referral**, **AI Job/Session**, **Support Ticket**.
- Provide an Entity-Relationship diagram or list of relationships (e.g., Merchant 1..* Stores, Store 1..* Products, Store 1..* Orders, User 1..* Orders, Order 1..* AI_Jobs, Merchant 1..* Wallet, etc.). [Mermaid ERD diagram to be added: show

Organization–>Merchants–>Stores–>Products/Orders/Invoices, Users–>Orders, Offers/Coupons linked to products or stores, etc.]

- For each entity, outline key attributes needed (ID, status, timestamps, etc.) and core state lifecycle (e.g., **Merchant**: applied → verified → active → suspended/frozen; **Store**: draft → listed → limited/hidden → delisted; **Product**: draft → active → out-of-stock/hidden; **Order**: created → pending → confirmed → fulfilled or cancelled/disputed; **Wallet**: active/above-minimum vs. frozen; **Invoice**: draft → sent → paid; **Offer/Coupon**: draft → active → expired; **Referral**: invited → joined → qualified → rewarded; **AI Job**: pending → generated → (retry), etc.).

2.2 Data and API Contracts (Schema)

- List the main data inputs needed from backend for each page (e.g., JSON fields). For example, **User List API** might return `[ { userId, name, phone, city, registeredAt, status, lastActive, totalOrders, totalSpent, tags, segment } ]`.
- Provide sample JSON schema snippets: e.g. `UserProfile { id, name, phone, email, joinedDate, lastSeen, scansCount, imagesGenerated, ordersCount, ticketsCount }`. Do similarly for **MerchantProfile**, **StoreDetails**, **ProductDetails**, **OrderDetails**, **WalletTransaction**, **Invoice**, **Offer**, **Coupon**, **ReferralStat**, **AIJobUsage**, **Ticket**.
- Map which page(s) will call which API and expect which data fields. For example: User Detail page requires userProfile, userOrders, userScans, userAIUsage; Merchant Detail page calls merchantProfile, merchantStores, storeAggregates, merchantWallet, merchantAnalytics.*
- Define event taxonomy: list core user events (login, scan, search, save, add-to-cart, book, purchase, refund), merchant events (list creation, update, order acceptance, wallet top-up), store events (store visited, store followed), AI events (chat start, image request, image retry), offer events (coupon applied, offer redeemed), etc. Map these events to pages: e.g. “booking_created” appears on Orders page, “image_generated” on AI usage page, “merchant_recharged” on Wallet page, etc. [Exportable table: Event → Page / Module].

2.3 Roles & Permissions (≈1,200 words)

- Define roles (Org Manager, Customer Support, Merchant Support, Operations Manager, Finance, Analytics, Tech Support).
- For each role, list allowed modules/actions. E.g., Customer Support can view Customers, Orders, Tickets, but cannot see merchant wallets; Finance can access Wallets, Invoices, Payment Gateway logs, but not necessarily individual tickets; Analytics can view all dashboards but not modify core data.*
- Provide a **Role Access Matrix** (table) mapping roles to pages or permissions (View/Edit). E.g.:

Role	Users	Merchants	Stores	Products	Orders	Wallet/Finance	Analytics	Support
Org Manager	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Customer Support	R	–	–	–	R	–	–	R/W
Merchant Support	–	R/W	R/W	R/W	R/W	R	–	R/W
Operations Manager	R/W	R/W	R/W	R/W	R/W	R	R/W	R
Finance	–	R	R	–	R	R/W	R	R
Analytics	R	R	R	R	R	R	R/W	R
Technical Support	R	R	R	R	R	R	R	R/W

"R" = Read, "W" = Write. (Adjust as needed.)

Explain any multi-tenant or account-level restrictions: e.g., a merchant admin can only see their own stores and orders. Org-level admins see all.

3. Navigation and Layout (≈1,200 words)

- **Left Navigation (persistent)**: Lists main modules as top-level links: Overview (Dashboard), Users, Merchants, Stores, Products, Orders, Wallet/Finance, Invoices, Offers/Coupons, Referrals, Analytics, AI Usage, Campaigns, Support Tickets, Trust & Safety, System Health, Audit Logs, Settings (Roles & Permissions). We follow common SaaS layout best practices.
- **Top Header**: Contains Org/Account selector (if multiple orgs), quick search (users, merchants, orders, tickets), notifications, user profile & logout. Possibly a date range picker for global date filter.
- **Breadcrumb/Title Area**: Under header, each page shows title and context (e.g., "Users > User Profile").
- **Content Layout**: Typically split-screen with filter sidebar (where applicable) or persistent table header filters, plus a main data table or forms. Multi-column cards or summary panels on overview pages. Responsive design can collapse nav.
- **Examples**: Amazon Seller Central uses a top nav of sections and a left sub-menu. GA4 uses a left nav with clear sections (Home, Reports, Admin). We will combine these patterns: e.g., left nav for major modules and second-level menu for subpages.

4. Page-by-Page Outline (≈15,000 words)

Below are 70+ pages (or logical page groups) with purpose, users, data/KPIs, UI layout sketch, controls, filters, tables, and states. For brevity we'll number them but group related pages. Each sub-section should include a one-line wireframe description and a button-behavior table.

4.1 Authentication & Access

4.1.1 Login / Role Selection Page

- **Purpose**: Entry to Admin Panel; user chooses/enters role if multi-role account.
- **Users/Roles**: All (Org Admins, Support, etc.).
- **Data Inputs**: Username/email, password, OTP (if 2FA). Possible role dropdown if user has multiple roles.
- **KPIs**: None.
- **UI Layout**: Centered login box with SimulaFly logo. Fields: [Email/Phone], [Password], [Role (dropdown or radio if applicable)], [Login button], [Forgot password link]. Use company branding.
- **Buttons/Controls**:
  - **Login** (primary): validates input, shows spinner on click, then navigates to dashboard.
  - **Forgot Password**: opens modal or page to reset.
  - **Role Dropdown**: auto-filled if user has multiple roles; disabled if only one.
- **States**:
  - Disabled **Login** until fields filled.
  - Error messages inline for invalid creds or 2FA.
  - Loading spinner on success.
  - If 2FA enabled, additional OTP field appears after correct password.
- **Permissions**: Unauthenticated (all).
- **Microcopy**: "Please enter your email or phone number." "Your password must be at least 8 characters."
- **One-line Wireframe**: Centered auth form with logo on top, fields and Login button

Button/Control	Label/Text	Action	Behavior
Login Button	"Login"	Submit credentials, navigate on success	Disabled until fields valid; shows spinner; on error shows message
Forgot Password Link	"Forgot password?"	Open reset flow (modal or page)	Opens modal with email input
Role Dropdown	"Select Role"	(if applicable) choose Org/Admin, etc.	Disabled if only one role; resets on logout

4.1.2 User Impersonation (Optional)

- **Purpose**: Allow Org Admin to impersonate a user/merchant (for support/debug).
- **Users**: Org Manager.
- **Data**: Search for user or merchant by name.
- **UI**: On top bar, Admin sees an "Impersonate" option. Clicking shows modal with search & select. Confirm triggers session as that user.
- **Controls**:
  - **Search Field** to find user.
  - **Impersonate Button** (confirmation).
- **States**: "You are now viewing as User XYZ" banner; "Exit Impersonation" link.
- **Microcopy**: "Type name or ID to find user."

Button	Label	Action	Behavior
Search/User Combo	"Find user/merchant"	Autocomplete filter list	Shows matching accounts as you type
Impersonate	"Impersonate"	Confirm and switch context	Disabled until account selected; shows confirmation toast
Exit (banner	"Exit	Revert to admin	Always visible while impersonating

link)	Impersonation"	account	

4.2 Overview Dashboard (Executive Command Center)

- **Purpose**: At-a-glance metrics and alerts for Org Managers (and Ops/Analytics teams).
- **Users**: Org Manager, Ops, Analytics.
- **Data/KPIs**:
  - **Commerce KPIs**: Total Bookings, Total Revenue, Active Merchants, Active Users, New Users today, Conversion Rate.
  - **AI KPIs**: Images Generated (total, per day), AI Spend (total cost), Cost per User Session.
  - **Support**: Open Tickets (consumer vs merchant), unresolved high-priority issues.
  - **Finance**: Wallet float (total merchant credit), failed payments.
  - Real-time Alerts (below).
- **Layout**:
  - Header with page title "Dashboard".
  - Top cards: small KPI boxes (with trend indicators).
  - Below cards: trend line charts (7-day bookings, bookings vs generated images, daily active users).
  - Alerts/flags: banner or side panel listing critical alerts (e.g., "12 Merchants below min balance", "3 payment failures in last hour", "5 urgent support tickets").
  - Quick Links: Buttons linking to high-priority queues (e.g. "View Low-Balance Merchants", "View Unconfirmed Orders").
- **Filters**: Global date picker (Today, 7d, 30d). Optional region or channel filters.
- **Controls/Buttons**:
  - Date Range selector (dropdown/calendar).
  - KPI cards may have "View Details" link.
  - Alerts have "Go to..." links.
- **Permissions**: Org Manager sees all; other roles see subset of KPIs relevant to them.
- **States**:
  - Loading charts skeleton, "No data" if new account.
- **Microcopy**: "Total bookings last 7 days", "Active merchants (30d)".
- **One-line Wireframe**: Top KPI cards and trend charts with alerts banner.

Button/Control	Label/Text	Action	Behavior
Date Range Picker	"Today / 7d / 30d / Custom"	Filter dashboard data by date range	Opens calendar for custom range; default Today
KPI Card Link	e.g., "View merchants"	Navigate to Merchants page	Visible on Org Managers only; disabled if no data
Alert Link	e.g., "Review ticket"	Jump to specific page or queue	Shows count; red badge for urgency
Refresh/Settings	Refresh icon	Refresh data	(if implemented)

4.3 Consumer Support (User Helpdesk)

This module is for app users (consumers) issues.

4.3.1 Consumer Support Ticket List

- **Purpose**: List all consumer support tickets for support agents.
- **Users**: Consumer Support, Org Manager.
- **Data**: Ticket ID, user reference (name/phone), issue category (app bug, image not showing, order issue, payment issue, general), status, priority, created date, last updated.
- **KPIs**: Number of tickets by category/status (display as totals above table).
- **Layout**: Table of tickets. Left filters: category, status (Open, Pending, Closed), date range, user. Main: list with columns [#][User][Category][Subject][Status][Last Updated][Assigned To]. Search box for user/TicketID.
- **Buttons/Controls**:
  - "View / Edit" on each row.
  - Filters (dropdowns, date pickers).
  - Bulk actions: "Assign", "Close", "Tag".
- **Statuses**: New, In Progress, Resolved, Closed. (State machine: new→ assigned→ resolved (customer ok)→ closed.)
- **States**: "Loading…", "No tickets found."
- **Permissions**: Support and Org Manager see all; others none.
- **Microcopy**: "Filter tickets by status or category."
- **Wireframe**: Sidebar with filters; main ticket table.

Button/Control	Label	Action	Behavior
Filter Dropdown	"Status: Open"	Filter table by status	Multi-select; default "All"
Search Box	"Search tickets…"	Search by user or Ticket ID	Shows suggestions; hits text input
View Ticket	"Open" link	Open ticket detail drawer/page	One click opens detail pane on right
Assign Button	"Assign"	Assign selected tickets to agent	Disabled if none selected; opens modal
Close Button	"Close"	Close selected tickets	Confirm dialog if multiple

4.3.2 Consumer Support Ticket Details

- **Purpose**: Show full context of a consumer's issue.
- **Users**: Consumer Support, Org Manager, (maybe Merchant Support if order issue).
- **Data**: Ticket fields (subject, description, category, user info, app version, device, last actions). Link to user's order or image generation if applicable. Comments log with timestamps.
- **KPIs**: N/A. Possibly SLA timer.
- **Layout**: Split panel or modal: Left: ticket meta (ID, category, status, created date, user profile summary). Center: conversation thread (chat with user or notes). Right: context (user recent activity, related order ID, image job status, wallet payments).
- **Buttons/Controls**:
  - **Reply** (if integrated with chat/email).
  - **Change Status** (e.g. "Resolve", "Close", "Reopen").
  - **Assign to Rep**.
  - **Escalate** (marks as urgent to Ops).
- **States**: Show "Closed" banner if ticket closed. Lock editing if user closed it.
- **Permissions**: Only Support team and Org managers.

- **Microcopy**: "Add internal note…" or "Reply to user…"
- **Wireframe**: Three-column layout: ticket details, conversation, user context.

Button/Control	Label	Action	Behavior
Reply/Note Box	"Write a response…"	Post reply or note	Initially empty; disabled if closed; supports attachments
Status Menu	"Change Status"	e.g., from "Open" to "Closed"	Confirmation on close; updates ticket
Assign Agent	"Assign To"	Assign to specific support rep	Dropdown of agents; default current
Link Order	"View Order"	Open related Order Detail page	Only if order ID present; disabled otherwise

4.4 Merchant Support (Merchant Helpdesk)

Module to handle merchant (store owner) issues about products, payments, wallets, and referrals.

4.4.1 Merchant Support Ticket List

- **Purpose**: Similar to consumer tickets but for merchants.
- **Users**: Merchant Support team, Org Manager.
- **Data**: Ticket ID, merchant name, category (catalog, wallet, referral, order flow, verification), status, created/updated.
- **KPIs**: Totals by category/status.
- **Layout**: Table with columns [#][Merchant][Category][Subject][Status][Last Updated] Filters on merchant, category, status, date.
- **Buttons/Controls**: "Open Ticket" detail, filters as above.
- **States/Permissions**: analogous to consumer tickets.
- **Wireframe**: Same as consumer ticket list, with "Merchant" column.

4.4.2 Merchant Support Ticket Details

- **Purpose**: View context of merchant issue.
- **Users**: Merchant Support, Org Manager.
- **Data**: Ticket fields, linked merchant profile, wallet summary, store count, order stats Show wallet balance and recent transactions, referral code status.
- **Layout**: Similar 3-columndetail: Ticket info, merchant profile summary, recent wallet & order activity.
- **Controls/Buttons**: "View Merchant Profile", "Adjust Wallet (+/-)", "Flag as Priority", change status.
- **States**: Show if merchant on hold, or if wallet below minimum (warning banner).
- **Wireframe**: Ticket convo with sidebar merchant info.

4.5 User Management (Consumers)

4.5.1 User Directory/List

- **Purpose**: Browse and search consumer accounts, see basic metrics.
- **Users**: Support, Analytics, Org.
- **Data**: User ID, name, phone/email, join date, city, status (active, dormant, churn), total scans, total images, total bookings, last active.

- **KPIs**: Counts of New, Active, Dormant users. Possibly filter by cohort.
- **Layout**: Table [#][Name][Phone][City][Registered][Status][DAU/MAU?][Bookings][Last Active]. Filters: date range (join date), status, city. Search by name/phone.
- **Buttons/Controls**: "View Profile".
- **States**: "Loading…" / "No users found."
- **Microcopy**: "Segment: [New, Returning, High Intent, etc]."
- **Wireframe**: User search box and table.

Button/Control	Label	Action	Behavior
Status Filter	"Status"	Filter by active/dormant/etc.	Multi-select
Search	"Search users…"	Query by name or phone	Autocomplete suggestions
View Profile	"View" icon	Open User Profile page	

4.5.2 User Profile Page

- **Purpose**: Show detailed user activity for support or analysis.
- **Users**: Support, Analytics, Org.
- **Data**: Personal info (name, contact), registration source (referral/organic), join date, status tags (e.g. "High Intent", "First-time Buyer"), last active. Activity metrics: number of rooms scanned, chats, images generated, products saved/carted, bookings, purchases, and refunds.
- **KPIs**: e.g., Active Days, Total Orders, Conversion % for this user.
- **Layout**: Top summary (profile card), followed by timeline charts or lists:
  - **Activity Timeline**: events (login, scan, chat, image, save, cart, booking).
  - **AI Usage**: number of images, chats (with button to view prompt logs).
  - **Orders**: list of bookings made, with status.
  - **Support Tickets**: any tickets raised by user.
- **Controls**: "Impersonate User" (Admin only), "Send Message" (if chat integration), "Add Note".
- **States**: If no activity, show "This user has not taken any actions yet."
- **Microcopy**: "Last active X days ago".
- **Wireframe**: Tabbed layout or vertical timeline view.

Button/Control	Label	Action	Behavior
Edit Tags	"Add Tag"	Tag user with segment	Dropdown of tags (New, High Intent, etc.)
Impersonate	"Switch to user"	Begin user session (if allowed)	Confirms admin password again
View Order	order link/icon	Go to that Order Detail page	In Orders list; highlights if urgent

4.6 Merchant Management

4.6.1 Merchant Directory/List

- **Purpose**: View all merchants on platform and their key stats.
- **Users**: Ops, Sales, Analytics, Org.
- **Data**: Merchant ID, name, signup date, verification status (pending/approved), account tier, number of stores, total orders, total revenue, wallet balance, account health (active/frozen).

- **KPIs**: Active vs. total merchants, pending approvals.
- **Layout**: Table: [#][Merchant][Stores][Orders][Revenue][Wallet][Status][Joined]. Filters: verification status, active/inactive, tier, region. Search by name or ID.
- **Buttons/Controls**: "View Profile", "Message Merchant". Bulk: "Send Notification".
- **States**: Show "Suspended" or "Verification Required" badge on rows.
- **Microcopy**: "Active stores / total stores".
- **Wireframe**: Search bar above a merchant table.

Button/Control	Label	Action	Behavior
Verification Filter	"All / Verified / Pending"	Filter merchants by verification status	
View Profile	"View"	Open Merchant Detail page	
Message	"Send Msg"	Open chat/email composer to merchant	Disabled if merchant not verified

4.6.2 Merchant Profile

- **Purpose**: Complete view of a single merchant's account.
- **Users**: Ops, Merchant Support, Finance, Org.
- **Data**: Merchant info (name, contact, GSTIN, documents), status, tier, date joined. Summary: #stores, #products, total orders, total revenue, wallet balance, referral code status, active promotions.
- **Sub-tabs**:
  - **Overview**: Summary as above, with performance chart (revenue over time).
  - **Stores**: List of their stores (link to Store detail).
  - **Orders**: All orders from this merchant's stores (or link to global orders filtered).
  - **Wallet**: Balance, recharge history, deductions (each line item with date, type).
  - **Coupons/Offers**: Merchant-specific discounts, referral invite stats.
  - **Support Tickets**: All tickets by this merchant.
  - **Invoices**: Merchant-level invoices (monthly statements).
  - **CRM / Contacts**: (if applicable) merchant's sales rep or contact person.
- **KPIs**: Active Store count, Conversion rate, Average booking value for merchant's stores.
- **Controls/Buttons**:
  - "Verify Account" (for Org Admin).
  - "Suspend" / "Reactivate".
  - "Credit Wallet" (Finance).
  - "View Store" (on each store row).
- **States**: If wallet below min, show warning; if KYC docs missing, banner alert.
- **Microcopy**: "GST Verified" tick or "Pending KYC".
- **Wireframe**: Tabbed panel with merchant info and metrics.

Button	Label	Action	Behavior
Verify Account	"Verify Merchant"	Approve KYC documents	Only visible if pending; requires confirmation
Suspend	"Suspend Account"	Temporarily disable merchant	Requires reason; sets status=Frozen
Credit Wallet	"Add ₹"	Manually add credit to wallet	Opens modal to enter amount
View Store	"View" (per	Open that Store's detail	

	row)	page	

4.7 Store Management

Each merchant can have one or more Stores. Stores may represent brand outlets or distribution centers.

4.7.1 Store Directory/List

- **Purpose**: Manage all store fronts under all merchants.
- **Users**: Ops, Merchant Support, Org.
- **Data**: Store ID, name, merchant, category, city/region, number of products, orders, revenue, listing status (draft, active, hidden, frozen, delisted), wallet usage.
- **Layout**: Table: [#][Store][Merchant][Category][City][Orders][Revenue][Status][Actions]. Filters: merchant, category, status, region. Search by store name.
- **Buttons/Controls**: "View Store", "Freeze/Unfreeze". Bulk: "Message Merchants".
- **States**: Indicate if store is "Hidden" or "Delisted" (grayed out).
- **Microcopy**: "X orders in last 30 days".
- **Wireframe**: Stores table with action icons.

Button	Label	Action	Behavior
Freeze Store	"Freeze"	Put store into limited mode	Changes listing state; disabled if store has active orders
Unfreeze	"Unfreeze"	Restore active listing	Only if previously Frozen
View Store	"View"	Open Store Profile	

4.7.2 Store Profile

- **Purpose**: Details and analytics for one store.
- **Users**: Ops, Sales, Merchant (for own stores), Org.
- **Data**: Store info (name, address, contact), status, merchant. Summary: #products, #orders, revenue, visits/impressions (if tracked), average order value, conversion.
- **Tabs/Panels**:
  - **Overview**: General info and metrics (traffic, conversion, sales).
  - **Products**: List of products in store (link to Product Detail).
  - **Orders**: Orders for this store.
  - **Customers**: Attribution of customers (including "imported" customers).
  - **Offers/Coupons**: Store-specific promotions.
  - **Wallet Impact**: Portion of merchant wallet spent on this store (if tracked).
  - **Invoices**: Store-level invoices (if split).
  - **Support**: Tickets for issues specific to this store.
- **KPIs**: Monthly sales vs. target, conversion rate, top-selling categories.
- **Controls**:
  - "Edit Listing" (change status, adjust priority/visibility).
  - "Highlight Products" (if featured).
  - "Close Store" (archive/delist).
- **States**: Banners if store is not meeting metrics or has unpaid fees.
- **Microcopy**: "Status: Active (visible to users)". "Listing State: Live".
- **Wireframe**: Tabbed interface like Merchant profile, focused on that store.

Button	Label	Action	Behavior
Edit Store	“Edit”	Edit store details (name, category)	Opens modal or page
Highlight/Feature	“Feature”	Mark as featured store	Toggles highlight badge
Delist/Close	“Delist Store”	Take store offline (soft delete)	Confirm “Store will be hidden”

4.8 Product Management

4.8.1 Product Directory/List

- **Purpose**: Browse all products across all stores.
- **Users**: Ops, Merchant Support, Analytics.
- **Data**: Product ID, name, store, merchant, category, price, stock status, listing state (active/hidden/out-of-stock), #images, likes/saves, #orders.
- **Layout**: Table: [#][Product][Store][Category][Price][Stock][Status][Orders]. Filters: merchant, store, category, price range, status. Search by product name/SKU.
- **Buttons/Controls**: “Edit”, “View Store”, “Disable/Enable”. Bulk: “Push Low Stock Alert”.
- **States**: Show “Out of Stock” or “Draft” tags.
- **Microcopy**: “Visible to users” vs. “Hidden”.
- **Wireframe**: Standard grid with product thumbnail column.

Button	Label	Action	Behavior
Edit	“Edit”	Open product edit form	
Disable	“Disable”	Hide product from catalog	Confirmation popup; toggles to Enable
View Store	“Go to Store”	Open the product’s store profile	

4.8.2 Product Profile / Edit Page

- **Purpose**: View/edit all details of a product.
- **Users**: Merchant (for own products), Ops, Catalog Admin.
- **Data**: All product metadata: title, description, images, price, variants, attributes, inventory count, SKU, tags, video. Related analytics: views, saves, add-to-cart, conversion rate, feedback score.
- **Sections**:
  - **Basic Info** (title, category, brand, description).
  - **Media** (images/videos).
  - **Pricing & Inventory** (price, discounts, stock).
  - **Attributes** (color, size).
  - **Visibility & Listing** (status toggles, SEO info).
  - **Analytics** (display chart of views/orders).
- **Controls/Buttons**:
  - “Save” (save draft), “Submit for Review”, “Publish/Unpublish”.
  - “Add Variant” (if multi-variant product).
  - “Delete Product”.
- **States**: “Draft” badge if not published. If unapproved, “Awaiting Approval”. If published, “Live”.

- **Permissions**: Merchants can edit own products; Org Admins can override.
- **Microcopy**: "Enter up to 500 chars." "Price must be > ₹1."
- **Wireframe**: Tabbed form layout.

Button	Label	Action	Behavior
Save/Publish	"Save" / "Publish"	Save product; go live if complete	Disabled until required fields filled
Submit	"Submit for Approval"	Send to review queue	Visible if Org oversight needed
Delete	"Delete"	Permanently remove product	Confirmation (if no orders exist)
Add Variant	"Add Variant"	Create a variant (size/color)	Opens variant form section

4.9 Orders & Bookings

4.9.1 Orders/Bookings List

- **Purpose**: Track all customer bookings and order states.
- **Users**: Ops, Merchant Support, Customer Support, Org.
- **Data**: Order ID, booking date, user, store/merchant, products, total value, booking fee, status (Pending, Confirmed, Completed, Cancelled, Lost, Disputed), assigned merchant, payment status (debited, refunded).
- **KPIs**: Counts by status, total revenue, average order value. Filters: date range, status merchant, user.
- **Layout**: Table: [#][Order ID][Date][Customer][Store][Products][Amount][Status][Action]. Filters on status, date, merchant.
- **Buttons/Controls**: "View Details", "Filter by state". Bulk actions: none typical.
- **Statuses/States**: Colored badges for state. "Pending" means waiting for merchant confirmation (no wallet charge yet). "Lost" for unconfirmed within window.
- **Microcopy**: Tooltip on booking fee "Charged on confirmation". "Pending since X hours".
- **Wireframe**: Orders table with status badges.

Button/Control	Label	Action	Behavior
Filter Dropdown	"State"	Multi-select order states	Select pending/confirmed/etc
Search	"Search orders…"	Search by order ID or user	
View Order	"View"	Open Order Detail page	

4.9.2 Order Detail / Timeline

- **Purpose**: Full record of an order/booking.
- **Users**: Ops, Support (both consumer and merchant), Merchant, Finance.
- **Data**: Customer info, store, merchant, products booked, images generated in room (if any), total value, booking fee, payment, order status timeline, actions (merchant confirm date, customer cancel, merchant cancel, dispute filed). Any dispute notes.
- **Layout**: Left summary (customer, store, totals), center timeline (events), right (payment & wallet info). Timeline shows: Created → (maybe) Merchant Accepted → Wallet Debited → Completed or Cancelled → Refund.
- **Controls**:

- **Order Actions**: "Mark Dispatched" (if manual), "Issue Refund", "Close Order".
- **Communication**: "Contact Merchant", "Contact Customer".

- **States**: Show red warning if delayed beyond SLA. If disputed, show "In Dispute" banner.
- **Microcopy**: "Dispute reason: ..."
- **Wireframe**: Order summary on top, vertical timeline below.

Button/Control	Label	Action	Behavior
Refund	"Initiate Refund"	Process a refund to user	Opens refund dialog (amount, reason)
Close Order	"Close"	Manually close (if needed)	Confirm (sets status=Completed)
Contact Party	"Message"	Open support chat with customer/merchant	Only visible if unresolved dispute

4.10 Wallet & Finance

4.10.1 Merchant Wallet Summary

- **Purpose**: Show merchant's wallet balance and transactions.
- **Users**: Finance, Merchant Support, Org. (Merchants see their own wallet in Merchant Portal).
- **Data**: Current balance, pending (if any), min required, transaction history (recharges booking fees, refunds, intel unlocks).
- **Layout**: Top: Balance card with threshold indicator. Below: table [Date][Type][Amount][Balance After][Notes]. Filters: date range, transaction type.
- **Transactions**: Type could be Top-up, Booking Fee Deduction, Refund, Penalty, etc.
- **Controls**: "Filter by type", "Export CSV".
- **States**: If below min, show "Account restricted" banner.
- **Microcopy**: "Minimum balance required is ₹XXX."
- **Wireframe**: Balance card and transaction table.

Button/Control	Label	Action	Behavior
Recharge	"Recharge"	Manually add funds	Opens recharge modal
Filter	"Type"	Multi-select trans. type	Dropdown list
Export	"Export CSV"	Download wallet ledger	Immediate CSV download

4.10.2 Company Financial Overview

- **Purpose**: Org-level finance dashboard.
- **Users**: Finance, Org Manager.
- **Data/KPIs**: Total wallet top-ups (revenues), booking fees collected, refunds issued, pending settlements, outstanding payments.
- **Layout**: Summary cards for inflow/outflow, chart of daily flows. Table of merchants below with balances.
- **Wireframe**: Finance KPI cards and merchant list.

4.11 Invoice Management

- **Purpose**: Generate and manage invoices to merchants for platform fees, and to companies for purchases (if applicable).
- **Users**: Finance, Org.

- **Data**: Invoice ID, merchant/store, period covered, line items (booking fees, subscriptions), amounts, tax. Status (draft, sent, paid).
- **Invoice Directory**: Table [#][Invoice][Merchant/Store][Period][Amount][Status]. Filters: merchant, date, status.
- **Invoice Detail**: Lists all charges, orders included, payments applied. Buttons: "Mark Paid", "Download PDF".
- **Merchant vs Store Invoices**: Possibly separate: e.g., one invoice per merchant per month listing all store charges. Also optionally per-store. Show hierarchy in Merchant profile.
- **States**: "Overdue", "Pending Payment".
- **Wireframe**: Invoice list and detail view.

Button	Label	Action	Behavior
Mark Paid	"Mark as Paid"	Record merchant payment for invoice	Confirmation; updates status
Download PDF	"Download"	Save invoice PDF	Opens PDF in new tab/download
Filter	"Status"	e.g., Unpaid / Paid	Multi-select

4.12 Offers and Coupons

4.12.1 Offer/Coupon List

- **Purpose**: Manage promotional offers.
- **Users**: Marketing, Sales, Org.
- **Data**: Offer/Coupon ID, code, description, type (SimulaFly vs Merchant), applicable scope (store/product/all), eligibility (all users or segments), discount value, validity, status (active/expired).
- **Layout**: Table: [#][Code][Type][Discount][Scope][Usage][ValidThru][Status]. Filters: active/inactive, type, merchant, store.
- **KPIs**: Number of active offers, redemptions, revenue from offers.
- **Controls**: "Create Offer/Coupon", "Edit", "Pause/Activate".
- **Rules**:
  - Merchant coupons vs global coupons.
  - Combinability rules (e.g., SimulaFly coupons may stack with merchant coupons or not).
  - Targeting (all users or specific user segment, first-time only, etc).
- **States**: "Usage limit reached", "Expired on XX".
- **Microcopy**: "* indicates one-time use."
- **Wireframe**: Promotions table with creation button.

Button	Label	Action	Behavior
Create Offer	"New Offer"	Open create-offer wizard	
Activate/Deactivate	"Activate" / "Pause"	Toggle offer	Confirmation needed if pausing active offer
Export	"Export"	Download usage report	CSV of redemptions

4.12.2 Offer/Coupon Details

- **Purpose**: Define or view a specific offer.
- **Users**: Marketing, Sales, Org.

- **Data**: All offer fields (code, % or flat off, min order, eligible users, applicable products/stores), redemption stats.
- **Layout**: Form with sections (General Info, Conditions, Usage Stats).
- **Controls**: "Save", "Cancel", "Delete".
- **Microcopy**: "Select specific users by uploading CSV."

4.13 Referral Program

- **Purpose**: Monitor and configure referral incentives.
- **Users**: Marketing, Ops.
- **Data**:
  - **Customer Referral**: invited customer phone, join date, status (pending, qualified), reward earned (₹50).
  - **Merchant Referral**: referrer merchant, referred merchant, status (registered, verified, activated), reward earned (₹500).
- **KPIs**: Total invites sent, conversions, total reward given vs pending.
- **Dashboard**: Summary cards for invites, joins, active conversions. Separate sections for customer and merchant referrals.
- **Tables**: Lists of referrals (columns: Referrer, Referee, Status, Date, Reward Released?).
- **Controls**: No typical buttons except possibly "Generate invite link", "Resend invite".
- **States**: "Pending reward", "Completed".
- **Microcopy**: "Reward releases after first booking by referred customer."
- **Wireframe**: Referral funnel chart + tables.

4.14 Prior-Share Attribution

- **Purpose**: Track how users were introduced (shares from WhatsApp, campaigns, merchants).
- **Users**: Marketing, Analytics.
- **Data**: Share events (who shared to whom, channel, timestamp), user join data, resulting activity.
- **Layout**: Graphs showing share counts by type, tree or funnel view.
- **Controls**: Filters by date, source type.
- **Wireframe**: Flow diagram of share → join.

4.15 Pricing Control Center

- **Purpose**: Configure all pricing parameters of the platform.
- **Users**: Org Manager, Finance, Ops.
- **Data**: Current booking fee structure (flat or %), AI usage cost per image/prompt, Buyer Intel price, referral bonus amounts.
- **Layout**: Form or table of rates. Each row: Item (e.g. "Booking Fee", "Image Credit"), Value, Effective Date, Action.
- **Controls**: "Edit", "Add Rule". Possibly tiered pricing (e.g. high-volume rate).
- **Rules**: Only Org Manager can modify. Changes logged.
- **Microcopy**: "Booking fee will be charged to customer's wallet upon merchant confirmation."

- **Wireframe**: Simple rate table with edit controls.

4.16 AI Operations Center

4.16.1 AI Usage Dashboard

- **Purpose**: Monitor generative AI usage and costs.
- **Users**: Org Manager, Analytics, Finance, Tech Support.
- **KPIs**: Images generated, Chat messages, Total token usage, Total cost, Cost per user/session. Breakdowns by model (if applicable).
- **Data**: Daily/weekly usage charts, usage by user/merchant, error/failure rates.
- **Layout**: Similar to analytics dashboard: cards + time-series charts. Filters: date range, by merchant or user segment.
- **Controls**: Date picker, filter by merchant or project (like [OpenAI's usage page][4]).
- **Wireframe**: Usage trend charts and breakdown tables.
- **Citations**: Use OpenAI usage UI as guide.

4.16.2 AI Job Log / Errors

- **Purpose**: View individual image/chat requests (for debugging).
- **Users**: Tech Support, Org.
- **Data**: Job ID, user ID, prompt text, status (success/failure), error message.
- **Layout**: Table or log.
- **Controls**: Retry button on failures.
- **Wireframe**: Table with "Retry" action on error rows.

4.17 Campaigns & Promotions

- **Purpose**: Manage marketing campaigns and targeted promotions.
- **Users**: Marketing, Sales.
- **Data**: Campaign ID, name, start/end date, budget, target group, coupons used, orders from campaign.
- **Layout**: List of active campaigns (with KPIs), campaign creation wizard.
- **Controls**: "Create Campaign", "Pause/Resume".
- **Wireframe**: Campaign list similar to Offers list.

4.18 Trust & Safety

4.18.1 Verification Queue

- **Purpose**: Manage merchant verification documents (PAN/GST, DigiLocker, etc.) and user flags.
- **Users**: Trust & Safety team, Org.
- **Data**: Pending verifications: merchant accounts with missing KYC, user accounts flagged for fraud.
- **Layout**: Tabs or lists for Merchant and User verification queues.
- **Controls**: "Approve", "Reject", "Request More Info". Bulk actions.
- **States**: "Verified", "Unverified".
- **Microcopy**: "All fields marked * are required."

- **Wireframe**: Simple queue interface with approve/reject buttons.

4.18.2 Suspensions & Appeals

- **Purpose**: Handle account suspensions or content violations.
- **Users**: Trust & Safety.
- **Data**: List of suspended users/merchants, reason codes, appeal status.
- **Controls**: "Reinstate", "Extend Suspension".
- **Wireframe**: Table of flagged accounts with actions.

4.19 Support & Escalations Center

- **Purpose**: Central view of support metrics and SLAs.
- **Users**: Support Managers, Org.
- **Data/KPIs**: Ticket resolution times, reopen rates, agent workload.
- **Layout**: Dashboards with charts and queue lists (e.g. "Tickets older than 24h", "Tickets by category").
- **Wireframe**: Charts and alerts for support SLAs.

4.20 System Health & Monitoring

- **Purpose**: Monitor integration and backend health (payment gateway, AI service, catalog sync).
- **Users**: Tech Ops, Org.
- **Data**: Uptime, failure rates, queue backlogs.
- **Layout**: Status panels, logs of recent incidents.
- **Wireframe**: Traffic light indicators for key systems.

4.21 Audit Logs

- **Purpose**: Record all admin actions for security/compliance.
- **Users**: Org, Security.
- **Data**: Timestamp, user/role, action type, target entity (e.g. "Frozen Merchant X"), old/new values.
- **Layout**: Table: [Date][Admin][Action][Object][Details]. Filters by action/user/date.
- **Controls**: "Export CSV".
- **Wireframe**: Basic log table with search.

4.22 Settings

4.22.1 Role Management

- **Purpose**: Define new roles, permissions.
- **Users**: Org (super-admin).
- **Layout**: Table of roles, list of permissions. Create/edit roles form.
- **Controls**: "Add Role", "Edit Permissions", "Delete Role" (if empty).
- **Wireframe**: Roles table + permission checkboxes.

4.22.2 Global Settings

- **Purpose**: Company settings (e.g. branding, currency, email templates).

- **Users**: Org.
- **Layout**: Various tabs for different settings categories.
- **Wireframe**: Form fields for company name, email, branding.

5. State Machines and Workflows (≈1,000 words)

Provide diagrams or bullet lists of state transitions for key entities:

- **Merchant**: Application → (verify) → Active → (low balance) → Limited → (further issues) → Suspended/Frozen → Reactivated. Actions: verify (Admin), recharge (Merchant/Auto), suspend (Admin).
- **Store**: Draft → Pending Approval → Active → Hidden/Limited (if flagged or low engagement) → Delisted → (possibly Reactivated). Triggers: merchant/ops updates metadata, admin reviews, fraud detection.
- **Product**: Draft → Submitted → Active → Hidden (if out-of-stock or policy issue) → Delisted/Archived.
- **Order**: Created (pending) → MerchantAccepted → WalletDebited → Completed or Cancelled. If merchant fails to accept in time → Lost. If user cancels → Cancelled. If dispute opened → Disputed → Resolved (refund or uphold). Roles: merchant, customer, admin can cancel.
- **Wallet**: BalanceAboveMin → (below threshold) → Freeze restrictions → Reactivate when topped up. Admin/finance can override.
- **Invoice**: Generated → Sent → Paid or Overdue. Can be adjusted or credited.
- **Offer/Coupon**: Draft → Active → Used/Expired.
- **Referral**: Invited → (joined) → Activated → Completed. Rewards: pending → released after conditions.
- **AI Job**: Pending → Success or Failed → (if failed) Retry or Cancel.
- **Ticket**: Open → InProgress → Resolved → Closed. Escalation raises priority.

(Mermaid state diagrams recommended for each above.)

6. Event Taxonomy and Analytics Ownership (≈1,000 words)

- **Event Categories**: define categories such as Acquisition (install, signup, referral accepted), Discovery (impression, search, category click, store open, product view, product save), AI Interaction (chat started, prompt message, image requested, image viewed, image liked), Conversion (add to cart, booking initiated, booking completed), Engagement (app open, session length), Off-platform (calls, external links).
-
- **Mapping to Pages**:
  - Pages like **Analytics Dashboard** will show aggregated metrics from Discovery (impressions, clicks), Conversion, Retention (repeat usage).

- **User Profile** & **Merchant Profile** show individualized events (appearance under "Activity").
- **Coupons/Offers pages** connect to events like coupon_redeemed.
- **Referral pages** map invites to signup events.

KPI to Page: Table listing each major KPI (DAU, MAU, bookings/day, conversion rate, CPC, wallet balance, AI cost) and which page shows it (e.g., DAU on Dashboard also on Analytics → Users).

Ownership: Specify that event logging (e.g. a user scanned a room) is done in-app and surfaced on Admin for **Analytics** or **Support** pages.

Provide an exportable table (as markdown or CSV snippet): e.g.:

Event	Triggers/Source	Shown on Page
user_signup	App signup screen	Users list, Acquisition report
room_scanned	App (consumer)	User profile, Analytics funnel
image_generated	AI service response	User profile, AI Usage page
product_saved	App UI (Save button)	User profile, Analytics
booking_started	End of cart flow	Orders list, Analytics funnel
booking_done	Order confirmed	Orders list, Dashboard, merchant analytics
referral_invite_sent	Web/App share	Referral dashboard
wallet_topup	Payment success	Wallet page, Finance summary
coupon_redeemed	Checkout	Offers page, Analytics
support_ticket_opened	Support UI	Support queue, Dashboard alerts
merchant_onboard	Merchant portal	Merchant list (pending)
...		

(Export the above table for copy/paste.)

7. State Machine Examples (with Mermaid Diagrams)

- Order Lifecycle:

```
Order Created
Merchant Confirms
Charge Wallet
Order Delivered
User Cancels
User Cancels (no fee)
No Merchant Response
Refund Issued
Customer Acknowledged
```

Pending
Accepted
Debited
Completed
Cancelled
Lost

```
Closed
```

Show code

- Referral Funnel:

```
Share or Invite
Referral Signup
First Action (scan/book)
Conditions Met
Reward Released
```

```
Invited
Joined
Activated
Rewarded
```

Show code

- Coupon Application:

```
mermaid
Copy
```

```
flowchart LR
    Checkout --\ "Enter Code" --> System
    System -- Valid? --> {Yes / No}
    Yes --> Apply Discount --> Order Total Adjusted
    No --> Show Error --> Return to Form
```

8. Wireframe Thumbnails (Sample Sketches)

(Provide simple thumbnail sketches for key pages: e.g., Dashboard, Merchant Profile, Order Detail, Analytics Funnel. Use markdown image embedding for small JPG/PNG if available.)
(Example: Wireframe of Merchant Profile page with tabs.)

9. Deliverable Checklist for UI/Frontend Teams

- **Design Comps**: High-fidelity mockups for each page listed above, annotated with requirements and responsive behavior.
- **Style Guide**: UI style definitions (colors, typography, icons).
- **Component Library**: Identify reusable components (tables, cards, modals, filters) needed.
- **Wireframes**: Low-fi wireframes for initial layout approval (for 5–10 key pages).
- **User Flows**: Diagrams for critical flows (order booking, wallet top-up, account verification).
- **Front-End Data Contracts**: JSON schemas / API contracts for each page's data input (as defined above).
- **Accessibility Review**: Ensure color contrast, keyboard navigation for all interactive elements.
- **Analytics Hooks**: List of data-layer events needed (from Event Taxonomy).
- **Role Testing Plan**: Scenarios to verify each role sees correct pages/buttons.
- **Export/Print Formats**: For invoice and report pages, define print/PDF layout.
- **Error States**: Designs for all error, empty, and loading states for each page.

By following this detailed, page-by-page outline (with roles, data, UI elements, and logic), the design and engineering teams will have a complete blueprint to build SimulaFly's Admin Panel front end. All decisions are informed by how established platforms structure their admin consoles, while covering SimulaFly's unique requirements (AI billing, referral economics, offline-customer attribution, etc.). This skeleton will be expanded into the full PRD (≈24k words) with flow diagrams, state tables, microcopy examples, and API field lists as needed.